LiTS + DeepSeek reproducibility

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- Question: how does maximum search depth affect accuracy and resource use?

- One fixed model; one fixed 30-question subset; three repeated runs.

Project and reproduction target

- Run the authors' MCTS workflow for numerical-answer MATH500 tasks.

- Policy proposes reasoning actions; transition builds a trajectory; reward scores it.

- Paper Table 6 is context: our model, reward setup and sample size differ.

Fixed source, data and model

- Pinned Git commit and Hugging Face dataset revision; SHA256 verified.

- CPython 3.12.12, API-only dependencies, DeepSeek Flash, thinking disabled.

- Per-request parameters, usage and model alias recorded; no keys in Git.

Controlled modification

- Maximum steps = 1, 3, 10; all groups enforce the same depth bound.

- Other controls: 10 MCTS iterations; branching 3; rollout 2; scoring fixed.

- Pilot informed the depth grid before formal experiments; overlap is disclosed.

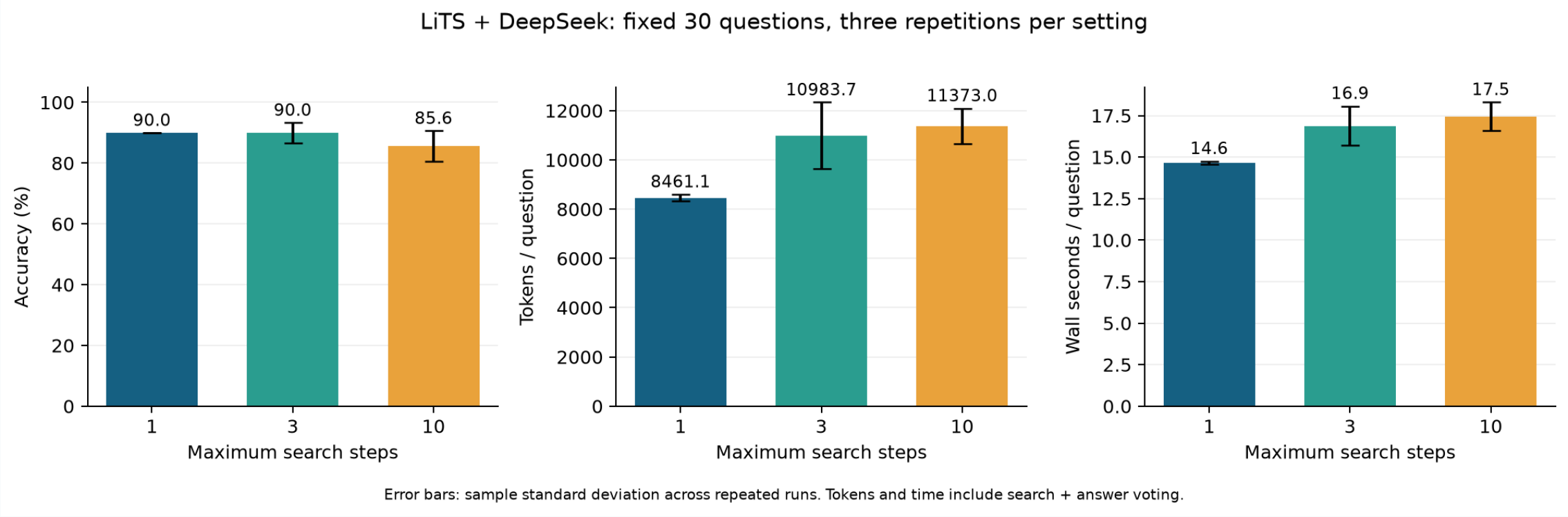
Measured results

- Depth 1: accuracy 90.0 +/- 0.0%; tokens/question 8461; seconds/question 14.6.

- Depth 3: accuracy 90.0 +/- 3.3%; tokens/question 10984; seconds/question 16.9.

- Depth 10: accuracy 85.6 +/- 5.1%; tokens/question 11373; seconds/question 17.5.

Accuracy, tokens and time



Why a step is not a token budget

- One generated action may contain a complete multi-operation derivation.

- The cap also changes the repository's final-step instruction.

- A larger external search budget does not guarantee better final answers.

Blockers and fixes

- Prepared an isolated compatible Python environment.

- Adapted DeepSeek thinking and stable tool-schema options.

- Fixed Windows UTF-8 artifact writing; preserved incomplete attempts.

Evidence and limitations

- Nine complete formal runs; 270 question-runs; only 30 distinct questions.

- Same-source fingerprints, trees, configs, numeric checker and raw usage saved.

- Strict scoring penalizes units/currency; small sample and pilot overlap limit conclusions.

Reproduce and conclude

- Default depth 10: 85.6 +/- 5.1% on this subset.

- Follow assignment/README_CN.md; inspect run_manifest and usage_summary.

- Submit the report, GitHub repository link and a Zoom recording under 15 minutes.